# Agentic AI – Student Design Handouts

These handouts are designed to guide you step by step through the final assignment: designing an Agentic AI system. Use the frameworks below to structure your thinking, justify your design choices, and clearly communicate how your agent works

## 1. Agent Goals

Define what success looks like for your agent. Be precise and outcome-oriented.

- Primary goal (one clear sentence)
- Why this goal matters (business or user value)
- Success metrics (how will you know the agent succeeded?)

What would be the Prompt for the Agent?

## 2. Input & Context

Describe what your agent observes and what information it needs to reason effectively.

- Triggers (what starts the agent?)
- Input data (tickets, alerts, user prompts, documents, telemetry, etc.)
- Context sources (knowledge bases, memory, historical data)
- What changes over time vs what stays static

## 3. Potential Tools & Actions

List the tools your agent can use and the actions it may take. Remember: tools are passive, agents decide when to use them.

- Tool name and purpose
- Type of action (read / write / execute)
- Required permissions
- Actions that require human approval

What would be the Tool description?

## 4. Agent Workflow (Perception → Reasoning → Action → Learning)

Explain how your agent operates over time using the agentic lifecycle.

- Perception: what the agent observes

• Reasoning: how decisions are made
• Action: what the agent does
• Learning: how feedback improves future behavior

## 5. Reasoning & Architecture Pattern

Choose the main design pattern that controls your agent and explain why.

Examples:
• ReAct (Reason + Act loop)
• Plan-and-Execute
• Multi-Agent (orchestrator, hierarchical, peer-to-peer)

Describe why this pattern fits your problem better than simpler automation.

## 6. Risks, Guardrails & Human-in-the-Loop

Identify what could go wrong(or extremely wrong) and how you prevent or mitigate failures.

• Potential risks (wrong action, hallucination, data misuse)
• Guardrails (constraints, validations, limits)
• Human-in-the-loop points (when the agent must pause or escalate)

## 7. Example Run (Walkthrough)

Provide a short, concrete example of your agent in action.

• Trigger
• Key reasoning steps
• Tool usage
• Final outcome

## 8. Resources and technology stack

Which LLM Model you would use, why?
Are you going to use any Cloud infrastructure provider to host your LLM model, why?
Programming language or/and platform to develop agents, Why?